

ADVANCE DATABASES

EXPERIMENT:02

Name:Chhavi Semwal

Sap:590017978

BCA/B4

EXPERIMENT:2.1

CODE:

```
1 > CREATE OR REPLACE FUNCTION GREATEST_OF_THREE (A,B,C)
5 RETURN VARCHAR2 IS
6     RESULT VARCHAR2(100);
7 BEGIN
8     IF
9         ( A > B )
10        AND ( A > C )
11    THEN
12        RESULT := 'The greatest number is A: ' || A;
13    ELSIF
14        ( B > A )
15        AND ( B > C )
16    THEN
17        RESULT := 'The greatest number is B: ' || B;
18    ELSIF
19        ( C > A )
20        AND ( C > B )
21    THEN
22        RESULT := 'The greatest number is C: ' || C;
23    ELSE
24        RESULT := 'Two or more numbers are equal and greatest.';
25    END IF;
26
27    RETURN RESULT;
28 END;
29 /
30
31 DECLARE
32     A NUMBER := &A;
33     B NUMBER := &B;
34     C NUMBER := &C;
35 BEGIN
36     DBMS_OUTPUT.PUT_LINE(GREATEST_OF_THREE(A, B, C));
37 END;
38 /
```

OUTPUT:

Substitution Variables

A

5

Cancel

OK

Substitution Variables

B

8

Cancel

OK

Substitution Variables

C

9

Cancel

OK

SQL> DECLARE

 A NUMBER := 5;

 B NUMBER := 8;

 C NUMBER := 9;...

Show more...

The greatest number is C: 9

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008

Experiment - 2

I Write a PL/SQL code to accept the values of A, b, c & display which is greater using functions.

```
CREATE OR REPLACE FUNCTION greater_of_three(a  
NUMBER, bNUMBER, cNUMBER)
```

```
RETURN VARCHAR2 IS
```

```
BEGIN
```

```
IF (a > b) AND (a > c) THEN
```

```
RETURN 'The greatest no is A: ' || a;
```

```
ELSE IF (b > a) AND (b > c) THEN
```

```
RETURN 'The greatest no is B: ' || b;
```

```
ELSE IF (c > a) AND (c > b) THEN
```

```
RETURN 'The greatest number is C: ' || c;
```

```
ELSE
```

```
RETURN 'Two or more numbers are  
equal & greatest';
```

```
END IF
```

```
END
```

```
/
```

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DECLARE

a NUMBER = 1 A;

b NUMBER = 1 B;

c NUMBER = 1 C;

BEGIN

DBMS_OUTPUT.PUT_LINE (greatest of three (a,b,c))

END;

/

Teacher's Signature : _____

EXPERIMENT:2.2

CODE:

```
1  CREATE OR REPLACE FUNCTION welcome_twenty
2  RETURN VARCHAR2 IS
3  |   msg VARCHAR2(4000) := '';
4  BEGIN
5  |   FOR i IN 1..20 LOOP
6  |   |   msg := msg || 'Welcome PL/SQL' || CHR(10);
7  |   END LOOP;
8  |   RETURN msg;
9  END;
10 /
11 BEGIN
12 |   DBMS_OUTPUT.PUT_LINE('Printing "Welcome PL/SQL" 20 times:' || CHR(10));
13 |   DBMS_OUTPUT.PUT_LINE(welcome_twenty);
14 END;
15 /
16
```

OUTPUT:

Printing "Welcome PL/SQL" 20 times:

```
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
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Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
Welcome PL/SQL
```

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2 Using functions display 'Welcome Pl/sql'
20 times ;

CREATE OR REPLACE FUNCTION welcome_twenty
RETURN VARCHAR2 IS

msg VARCHAR2(4000) := '' ;

BEGIN

FOR i IN 1..20 LOOP

msg := msg || TO_CHAR(i) ||
Welcome Pl/sql ||

END LOOP;

RETURN msg ;

END;

/

BEGIN

DBMS_OUTPUT.PUT_LINE ('Printing "Welcome Pl/sql'
20 times' || CHR(10));

DBMS_OUTPUT.PUT_LINE (welcome_twenty);

END;

/

EXPERIMENT:2.3

CODE:

```
CREATE OR REPLACE FUNCTION factorial(n NUMBER)
RETURN VARCHAR2 IS
|   fact NUMBER := 1;
|   result VARCHAR2(100);
BEGIN
|   IF n < 0 THEN
|       RETURN 'Factorial is not defined for negative numbers.';
|   ELSIF n = 0 THEN
|       RETURN 'Factorial of 0 is 1';
|   ELSE
|       FOR i IN 1..n LOOP
|           fact := fact * i;
|       END LOOP;
|       result := 'Factorial of ' || n || ' is ' || fact;
|       RETURN result;
|   END IF;
END;
/
DECLARE
|   num NUMBER := &Enter_a_number;
BEGIN
|   DBMS_OUTPUT.PUT_LINE(factorial(num));
END;
/
```

OUTPUT:

Substitution Variables

Enter_a_number
15

Cancel

OK

Factorial of 15 is 1307674368000

PL/SQL procedure successfully completed.

3 Write a PL/SQL code to find a factorial of a no using functions.

CREATE OR REPLACE FUNCTION FACTORIAL(n NUMBER)
RETURN VARCHAR IS

FACT NUMBER := 1;

BEGIN

IF n = 0 THEN

RETURN 'Factorial is not defined';

ELSE IF n = 0 THEN

RETURN 'Factorial of 0 is 1';

ELSE

FOR i IN 1..n LOOP

FACT = FACT * i;

END LOOP

RETURN 'Factorial of '|| n || ' is ' ||
fact;

END IF;

END;

/

DECLARE

NUMBER := 1 Enter a number;

BEGIN

DBMS_OUTPUT.PUT_LINE (factorial (num));

END;

/

EXPERMENT:2.4

CODE:

```
1  CREATE OR REPLACE FUNCTION fibonacci_series(n NUMBER)
2  RETURN VARCHAR2 IS
3      a NUMBER := 0;
4      b NUMBER := 1;
5      c NUMBER;
6      result VARCHAR2(4000) := '';
7  BEGIN
8      IF n <= 0 THEN
9          RETURN 'Please enter a positive number.';
10     ELSIF n = 1 THEN
11         RETURN 'Fibonacci series: ' || a;
12     ELSE
13         result := 'Fibonacci series: ' || a || ', ' || b;
14         FOR i IN 3..n LOOP
15             c := a + b;
16             result := result || ', ' || c;
17             a := b;
18             b := c;
19         END LOOP;
20         RETURN result;
21     END IF;
22 END;
23 /
24 DECLARE
25     num NUMBER := &Number_of_terms;
26 BEGIN
27     DBMS_OUTPUT.PUT_LINE(fibonacci_series(num));
28 END;
29 /
30
```

OUTPUT:

Substitution Variables

Number_of_terms

5

Cancel

OK

```
SQL> DECLARE
      num NUMBER := 5;
      BEGIN
          DBMS_OUTPUT.PUT_LINE(fibonacci_series(num));...
Show more...
```

Fibonacci series: 0, 1, 1, 2, 3

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.004

4 Write a PL/SQL code to generate fibonacci series using functions.

CREATE OR REPLACE fibonacci_series
(n NUMBER)

RETURN VARCHAR2 IS

a NUMBER := 0;

b NUMBER := 1;

c NUMBER a;

result VARCHAR2(4000) := ' ';

BEGIN

IF n < 0 THEN

RETURN 'Please enter a positive number.

ELSE IF n = 1 THEN

RETURN 'fibonacci series ' || a;

ELSE

result := fibonacci_series || a || ' ';

FOR i IN 3..n LOOP

c := a + b;

result := result || ' ' || c;

a := b;

b := c;

END LOOP;

RETURN result;

END IF;

END;

/

Teacher's Signature : _____

EXPERIMENT:2.5

CODE:

```
CREATE OR REPLACE FUNCTION sum_first_n(n NUMBER)
RETURN VARCHAR2 IS
    total NUMBER := 0;
BEGIN
    IF n <= 0 THEN
        RETURN 'Enter a positive number.';
    ELSE
        FOR i IN 1..n LOOP
            total := total + i;
        END LOOP;
        RETURN 'Sum of first ' || n || ' natural numbers is: ' || total;
    END IF;
END;
/

DECLARE
    num NUMBER := &Enter_a_number;
BEGIN
    DBMS_OUTPUT.PUT_LINE(sum_first_n(num));
END;
/
```

OUTPUT:

Substitution Variables

Enter_a_number

4

Cancel

OK

Sum of first 4 natural numbers is: 10

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005